Luna Final Report - Pranav Manoj

Introduction:

Stars including our sun are constantly producing immense amounts of energy through nuclear fusion. The source of this energy is given by the binding energy of atomic nuclei, which can be released in particular when converting protons (zero binding energy) up to medium-mass nuclei. In order to properly understand stellar structure and evolution, it is thus necessary to understand how light nuclei are converted to heavier ones.

Nuclear reactions in stars synthesize chemical elements. All but the lightest five elements (hydrogen to boron) are predominantly produced in stars, and for example the crucial ratio of carbon to oxygen abundances is controlled by stellar helium burning. These thermonuclear reactions shape the life and death of stars and are responsible for the synthesis of the chemical elements in the Universe. To better understand these reactions and thus, better understand stellar evolution and structure, it is important to measure the thermonuclear reaction rate (cross-sections) for all the nuclear processes involved. Measuring these nuclear cross-sections is very challenging and is the main goal of experimental nuclear astrophysics.

In stellar environments, the energy of the interacting particles is usually much lower than the Coulomb repulsion between the nuclei. Under these conditions, the nuclear reactions can only proceed via the quantum tunneling effect. As a result, the reaction cross sections at relevant energies can be extremely low, and very difficult to measure since the signal can be lost in the natural background. A turning point in this context has been the introduction of underground measurements performed for the first time at the Laboratory for Underground Nuclear Astrophysics (LUNA) laboratory located in Gran Sasso, Italy. The deep location of LUNA, which is shielded by 1400 meters of dolomite rock from cosmic ray induced effects, optimizes sensibly the signal-to-noise ratio for the nuclear reactions at astrophysical energies.

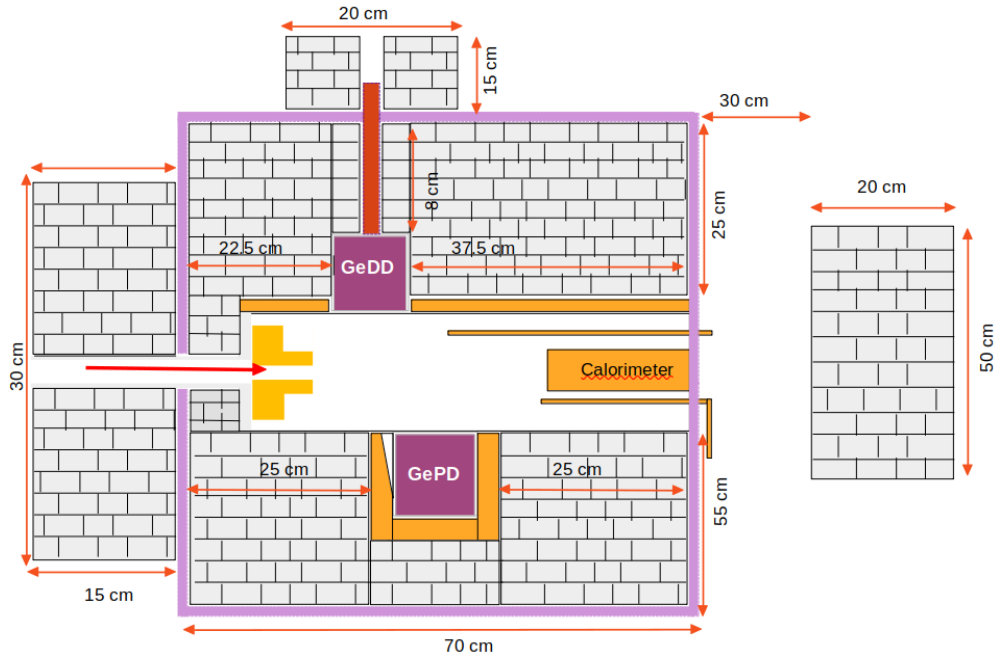
Among the cycles of nuclear reactions which contribute to the synthesis of chemical elements inside stars, the NeNa cycle has a very crucial role in the nucleosynthesis of elements from ^{20}Ne to ^{24}Mg in several stellar environments such as in red giants stars (during H shell-burning), asymptotic giant branch stars, novae, and massive stars. The NeNa cycle begins with ^{20}Ne which can be produced in stellar sites through several phases of burning. Starting with proton captures on Ne isotopes, all NeNa reactions are activated in stellar scenarios characterized by different temperature ranges of interest. To constrain the role of the NeNa cycle, a precise knowledge of the reaction rates of all nuclear reactions involved is required.

In particular, the ^{20}Ne(p,γ)^{21}NaNa reaction is the bottleneck of the cycle, being the slowest one. However, in the literature, there are only a few measurements of its cross section using both direct and indirect approaches. All these studies cover only the energy region above 370 keV. For these reasons, a new measurement is ongoing at LUNA and my work over the past 2 months was focused on this study

Experimental Setup:

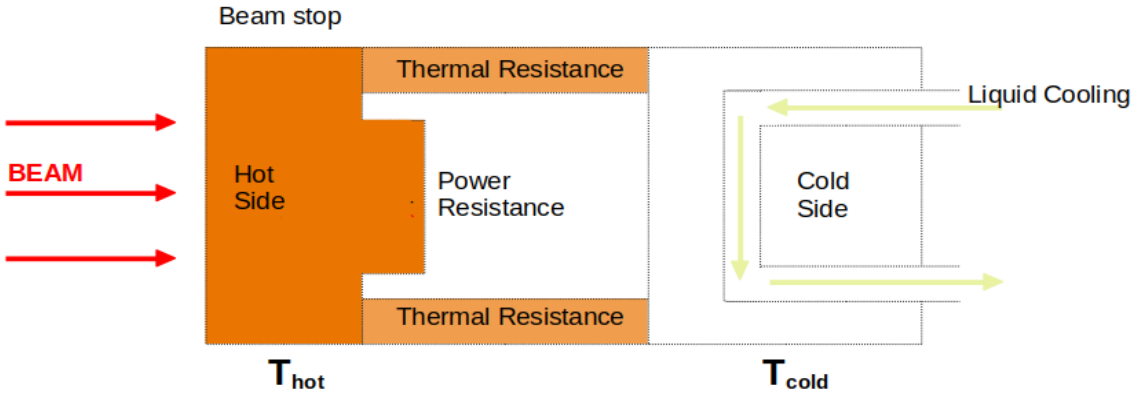
The measurements of the ^{20}Ne(p,γ)^{21}Na reaction were taken using the LUNA 400 keV accelerator combined with a high-resolution HPGe detection system. The setup used is illustrated below. It consists of two high-purity germanium detectors: an Ortec low background detector with 90% relative efficiency (shown here also as GeDD) and a Canberra low background detector with 130% relative efficiency (shown here also as GePD). The GePD detector, located below the interaction chamber at 13.9 cm from the beam collimator, is surrounded by a 4 cm thick copper shielding and a 25 cm thick lead shielding. The copper shielding is needed to reduce the bremsstrahlung γ-rays produced by the interaction of the electrons emitted in the β-decay of

210 Bi inside the lead shielding. The GeDD detector, located on top of the interaction chamber at 7.03 cm from the beam collimator, is surrounded only by the lead shielding. Between the lead shielding above the chamber and the chamber, a 3 cm thick copper shielding is used.



The setup enclosed in the lead castle is itself enclosed in a plexiglass radon box shown by the violet line in the figure. The radon box establishes an N_2 overpressure around the lead castle and avoids radon gas flushing inside. It also prevents radon gas build-up around the detectors flushing inside the castle the nitrogen exhaust from the GePD detector dewar. Moreover, to avoid the γ-rays produced from the interaction of the beam with the beam line collimator, a 15 cm lead shielding between the pumping first stage and the chamber is used. Instead, on the right side, the external lead wall located behind the chamber prevents the detectors from the γ-rays which can reach the chamber through the end-flange.

For the measurements of the $^{20}Ne(p,\gamma)^{21}Na$ reaction, two independent data acquisition systems are adopted: CAEN digital acquisition and the analog multichannel acquisition. The CAEN digital acquisition allows the storage of the single events recorded by the two detectors in list mode configuration. For each event, timestamp, energy, and pileup are saved and can be sorted offline according to the required analysis.



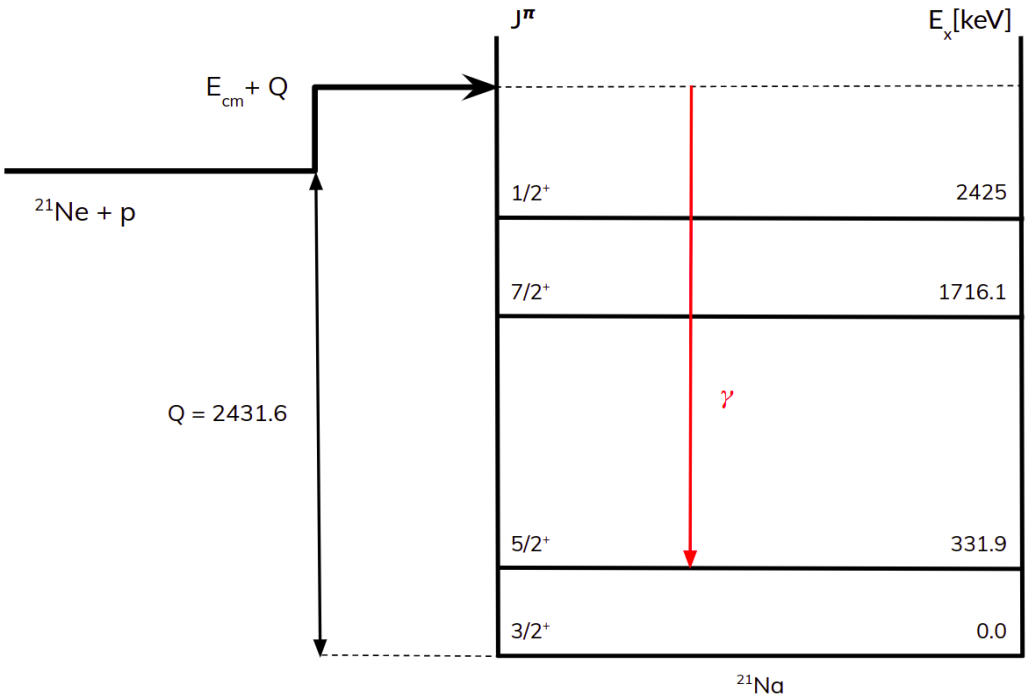
To measure the beam current a calorimetric approach is adopted. The beam stop is a calorimeter made up of a hot side , and a cold side in thermal contact to each other. The current intensity is measured using a constant temperature gradient calorimeter. A cooling machine keeps the cold side of the calorimeter at a constant temperature called T_{cold} (usually $T_{cold} = -5^{\circ}C$), while the hot side temperature is adjusted by heating resistors. A feedback system reads the hot side temperature through PT100 thermoresistors and adjusts the resistors power (the so-called zero power, W_0) to keep the beam stop at a constant temperature T_{hot} (usually $T_{hot} = 70^{\circ} - 75^{\circ}$ C). The W_0 depends on the gas type and pressure on the chamber. When the beam impinges on the hot side, the latter is heated up both by the beam and the thermoresistors. In this case, the power provided by the resistors to keep the hot side at T hot will be lower. Measurements of the resistor's powers can be then used to calculate the beam intensity.

Analysis:

The primary reaction which I studied was the radiative direct capture of the $^{20}Ne(p,\gamma)^{21}Na$ reaction. This process is characterized by the complete absorption of the incident projectile by the target and immediate ejection of one or more photons with energy E_γ:

$$E_\gamma = E_{cm} + Q_{value} - E_i$$

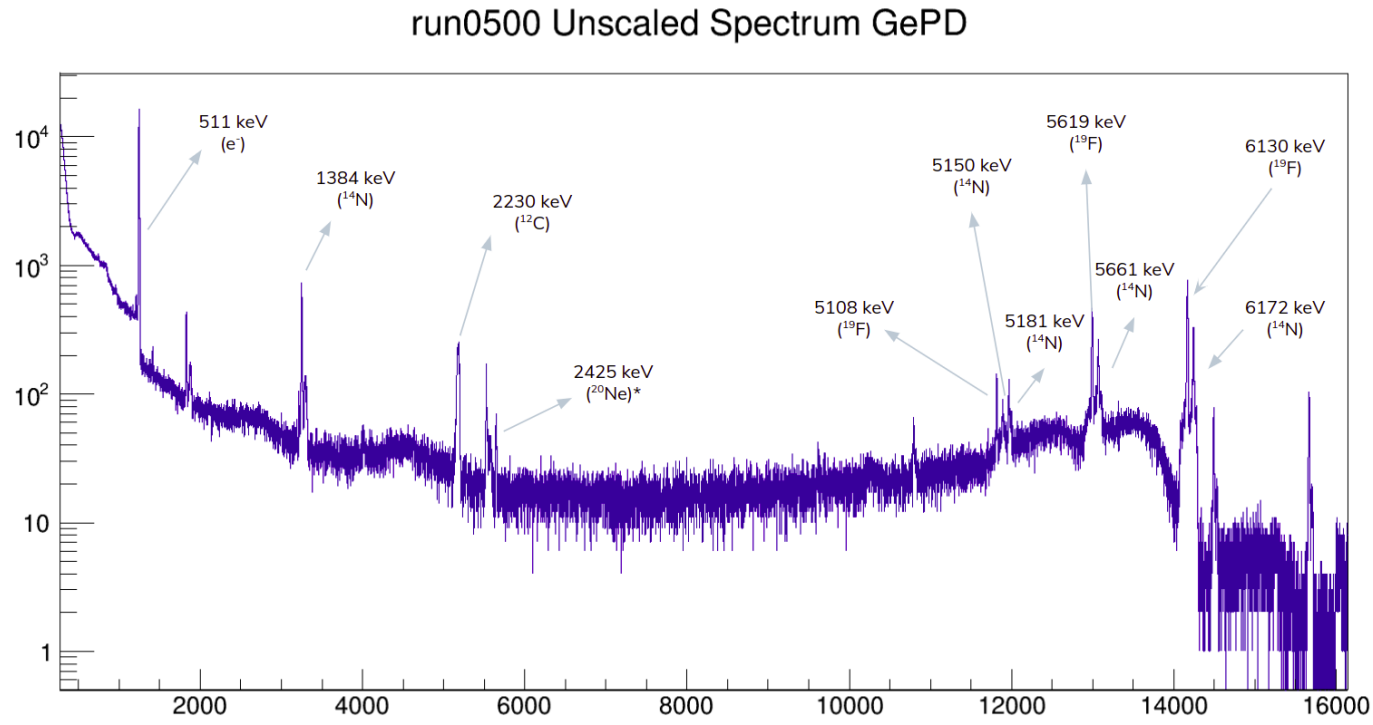
where Q_{value} is the energy released in the reaction, E_{cm} is the proton beam energy in the center of mass frame, and E_i is the energy of the level populated in the target nucleus. The ejected photons are strongly peaked along the trajectory of the incident projectile. The figure below shows a schematic representation of the direct capture mechanism. The direct capture is a purely electromagnetic process which dominates at energies above 20 MeV, or when there are no strong resonances in or near the Gamow window. It is also particularly important at low energies in light nuclei reactions where the density of resonances is very low, as in the present case.



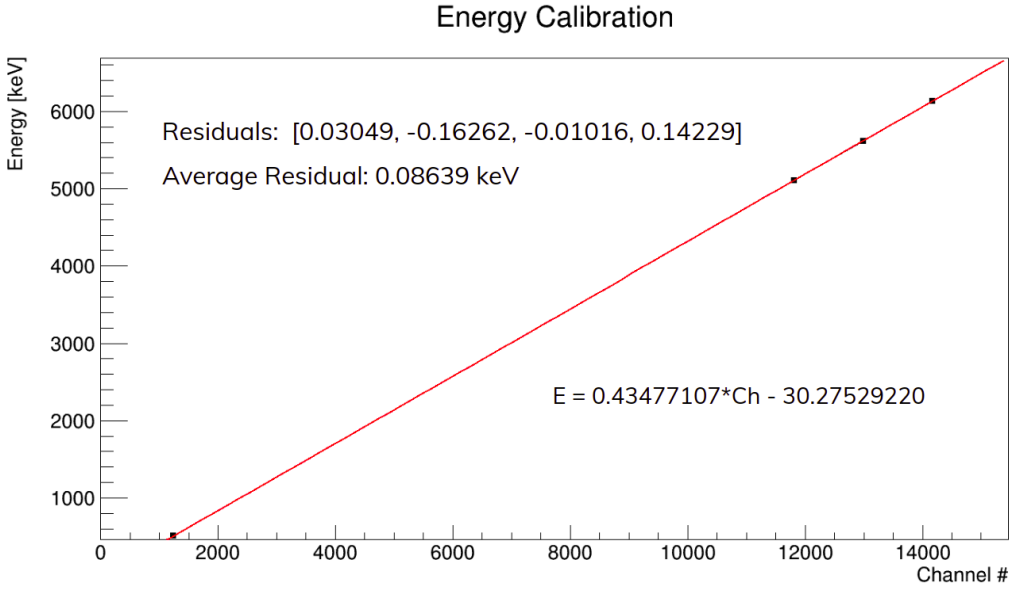
Nuclear reactions can also proceed through resonant mechanisms; in this case the excited state E_r of the compound nucleus is populated in the entrance channel and then decays to a lower state. This process happens only when the energy $Q_{value} + E_{cm}$ of the entrance channel coincides with the energy of the excited state in the compound nucleus:

$$E = E_r = Q_{value} + E_{cm}$$

The gamma rays released from the transitions either through the direct capture or resonant mechanism can be seen as peaks in the recorded spectrum. The figure below is an example of an uncalibrated spectrum where the horizontal axis is measured in channels and the vertical axis is measured in counts.



After arriving in Italy and learning about the LUNA experiment, one of my first tasks was to calibrate the above spectrum using the background peaks from known transitions which are labeled in the above spectrum. Using the energies of these peaks, it is possible to find a linear calibration, $E = m*ch + b$, where m and b are parameters.

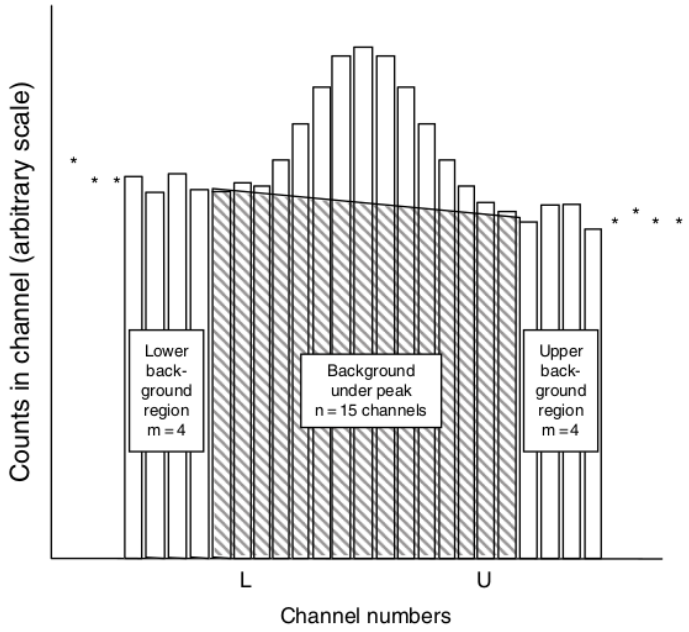


The energy calibration can change if for instance the gain of the spectroscopy amplifier of the HPGe detectors is modified, as well as other factors in the experimental setup. Therefore, I had to calibrate multiple spectra in order to ensure an accurate energy calibration especially in the relevant energy range. The accuracy of the calibration can be checked using the residuals of the linear fit; for our purposes, residuals under 1keV were required. An example of such an energy calibration is shown above.

I then had the opportunity to go to the underground laboratory at Gran Sasso. There, I was able to learn about the data taking procedure by performing a resonance scan of the $^{21}Ne(p,\gamma)^{22}Na$ reaction belonging to the NeNa cycle. The resonance scan involved taking multiple measurements at different beam energies (beam energy is changed by adjusting terminal and probe voltages of the accelerator) to determine the energies at which the gamma spectrum peaks from the E_p = 272 keV resonance gave the highest yield. The yield of a particular peak is calculated as a ratio of counts to accumulated charge.

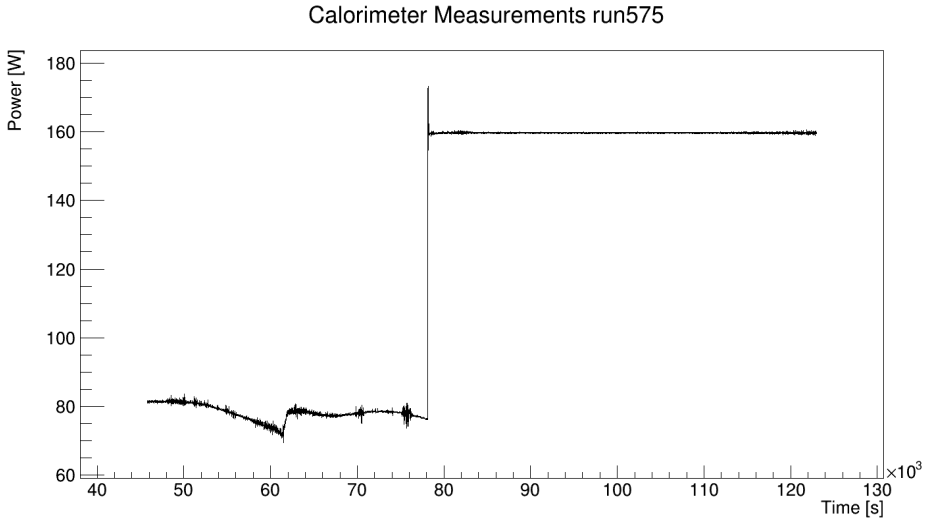
Since the two HPGe detectors are located at different positions along the target chamber, the beam energy which corresponds to the maximum yield is different for the two detectors. In order to calculate yield, the number of counts in the peaks of interest must be calculated. For this, the method I used was the Gilmore method. This method involves taking the integral of the region of interest and subtracting the background counts as estimated using the background regions at the right and left of the peak.

$$A = \sum_{i=\mathrm{L}}^{\mathrm{U}} C_i - n\left(\sum_{i=\mathrm{L}-m}^{\mathrm{L}-1} C_i + \sum_{i=\mathrm{U}+1}^{\mathrm{U}+m} C_i\right)/2m$$



Returning to the data analysis of the $^{20}Ne(p,\gamma)^{21}Na$ reaction, the next step, after properly calibrating all the relevant spectra, was to check the calorimeter data from each run to ensure that the beam current was stable during the run. Graphing the calorimeter power vs time allowed me to see if there were any problems during the measurement. There were problems in a few of the runs due to a sudden shut down of the accelerator terminal voltage during the run. An example is shown below. Problematic runs had to be adjusted by cutting off the data taken after the beam went off by using partial spectra saved each hour. Only after cutting out problematic data could

these spectra be taken into consideration. In addition, taking the average power during the run allows us to calculate the beam intensity.



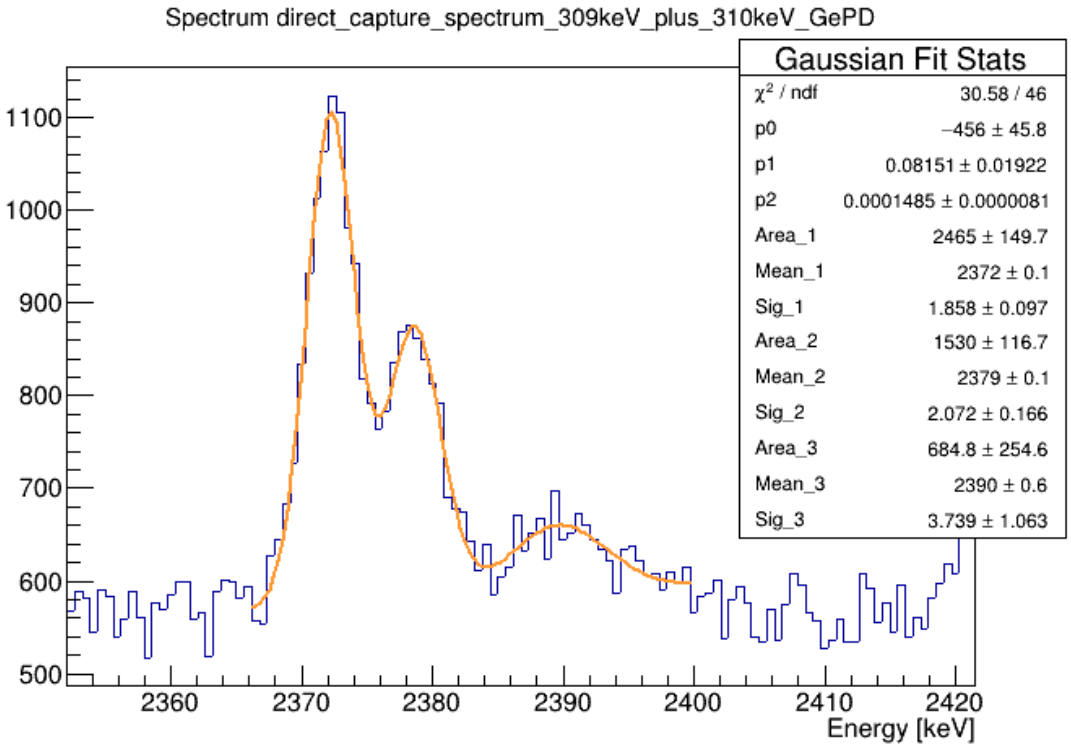
After using the calorimeter data to fix the problematic runs, I was then able to sum up the spectra at similar ion beam energies in order to increase the statistics. In order to sum up the spectra accurately, it was especially important to have an accurate energy calibration for each spectra so that the peaks of the spectra line up. Once the spectra were added up, it was possible to start analyzing the data and search for the expected transitions of the $^{20}Ne(p,\gamma)^{21}Na$ reaction. Calculating the gamma energies of each transition required me to take into account multiple factors such as the target density entering the calculation of the beam energy loss for each detector. In this estimate I have also considered the beam heating effect, which reduces the effective neon gas density due to the heating by the incident proton beam. The final list of expected direct capture primary and secondary transitions for each ion beam energy are shown below. In addition to the beam energies shown below, the 309.3 and 310.0 keV spectra were added together separately, since the two energies are very close by.

Ep[keV]	Run Numbers	E_y [keV] GePD Primary				E_y [keV] GeDD Primary			
260.1	474, 475, 582, 584, 586, 627	246.0	954.9	2339.1	2671.0	250.1	959.0	2343.2	2675.1
263.2	565, 567, 568, 569, 575	245.5	954.4	2338.6	2670.5	251.1	960.0	2344.2	2676.1
265.0	566	247.5	956.4	2340.6	2672.5	253.0	961.9	2346.1	2678.0
309.3	576, 577, 593, 622, 623, 624, 625, 626	293.1	1002.0	2386.2	2718.1	297.0	1005.9	2390.1	2722.0
310.0	500, 589, 628, 629, 630, 477	293.9	1002.8	2387.0	2718.9	297.7	1006.6	2390.8	2722.7

Egamma [keV] Secondary			
708.9	1384.2	331.9	
2093.1	1716.1		
2425			

In order to arrive at the cross section, it is necessary to calculate the areas of each of the expected peaks. To do this, I used two methods: the Gilmore method and the Gaussian fitting method. The Gilmore method (described earlier) works well for well defined peaks which do not overlap with other peaks. However, for peaks that overlap or have definite shapes, the fitting method is required. In the case that a peak overlaps with another background peak, the

contribution of this overlapping peak must be subtracted. This can be achieved by fitting the spectra with multiple normalized Gaussians where the area of the peak is represented by the first fit parameter. In addition, a third second degree polynomial is added to the fitting function to estimate the background. An example of this procedure is shown below for a primary peak. I am currently working on refining this procedure to get more accurate fits for peaks that have a less gaussian shape.



Conclusion:

This ongoing analysis is crucial for the determination of the $^{20}Ne(p,\gamma)^{21}Na$ cross section, which in turn is essential to the study of nucleosynthesis in the NeNa cycle in several astrophysical scenarios. This reaction is very slow acting as the bottleneck of the cycle. There are very few measurements of its cross section, which is why this research is crucial for the understanding of nucleosynthesis in many stellar environments. Furthermore, I have learned a great deal about nuclear astrophysics and the experimental techniques that are used to study relevant nuclear reactions. I was able to see the data taking process which helped in understanding the setup of the experiment. I also gained experience with data analysis by analyzing gamma ray spectra. There are many factors to take into account when analyzing data such as energy calibration, beam energy loss, calorimeter data, the beam heating effect, etc. The goal of analyzing the spectra was to identify the relevant transitions and calculate the peak area. To do this, I learned how to implement the Gilmore method and the fitting method. Overall, the experience I have gained while contributing to important measurements of the $^{20}Ne(p,\gamma)^{21}Na$ reaction has been invaluable.

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